

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199.

Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

CODE:

```
GNU nano 7.2 sstf.c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int req[] = {98,183,41,122,14,124,65,67};
    int n = 8;
    int head = 53;
    int visited[8] = {0};
    int total = 0, i;

    for(i = 0; i < n; i++) {
        int min = 9999, pos = -1;

        for(int j = 0; j < n; j++) {
            if(!visited[j]) {
                int dist = abs(head - req[j]);
                if(dist < min) {
                    min = dist;
                    pos = j;
                }
            }
        }

        total += min;
        head = req[pos];
        visited[pos] = 1;
    }

    printf("Total Head Movement = %d\n", total);
    return 0;
}
```

OUTPUT:

```
laxmi@laxmi:~$ nano sstf.c
laxmi@laxmi:~$ gcc sstf.c -o sstf
gcc: error: unrecognized command-line option '-o'
laxmi@laxmi:~$ gcc sstf.c -o sstf
laxmi@laxmi:~$ ./sstf
Total Head Movement = 323
laxmi@laxmi:~$
```